Abstract citation ID: keaf142.288

E052 EVALUATING REAL-WORLD USE AND EFFICACY OF TERIPARATIDE IN OSTEOPOROSIS MANAGEMENT

Roshni Vinda, Manahil Afzal, Muhammad K. Nisar

Rheumatology, Luton & Dunstable University Hospital, Luton, United Kingdom

Background/Aims

Teriparatide, a recombinant human parathyroid hormone, significantly reduces the risk of fractures in patients with postmenopausal and glucocorticoid-induced osteoporosis. In the UK, NICE has restricted its use to high-risk patient groups. This project aims to analyse the use of teriparatide at a large University teaching hospital with 55% ethnic catchment population, providing insight into the real-world patient demographic and evaluating the treatment's effectiveness.

Methods

A retrospective analysis was conducted to evaluate the use of teriparatide in patients receiving treatment between 2014-2024 at our unit. Full demographic and clinical data was collected using electronic health records and a live multipurpose database. Dual-energy X-ray absorptiometry (DXA) scan results were reviewed to assess bone mineral density (BMD) both before the initiation of Teriparatide treatment and after completion. Data was analysed to provide insights into the demographic characteristics of the treated population, the efficacy of teriparatide based on BMD improvements, treatment adherence, and local compliance with national guidelines.

Results

A total of 486 patients were prescribed teriparatide over the ten-year study period. This data is a snapshot from the first 54 consecutive patients, all women with an average age of 72.7 years. Of these, 52 (96%) were White and 2 (4%) were Asian-Indian. Alendronate was the most common first-line bisphosphonate. On average, patients experienced two fractures before starting teriparatide. All patients had failed previous treatment with at least one bisphosphonate or Denosumab, averaging 12.6 years of prior treatment. Nine patients (17%) discontinued treatment early due to side effects. Pre-treatment, the average BMD was 0.771 g/cm^2 for the lumbar spine (L1-4) and 0.646 g/cm^2 for the left hip. Post-treatment, these values improved to 0.860 g/cm^2 and 0.684 g/cm^2, respectively. The mean lumbar spine T-score improved from -3.3 to -2.6, and the Left hip T-score from -2.8 to -2.6. After teriparatide, 40 patients (74%) transitioned to denosumab, 7 (13%) began zoledronic acid, 5 passed away, and 2 were lost to follow-up.

Conclusion

Our study demonstrates real world use of teriparatide in a multi-ethnic diverse setting. It shows that teriparatide is used predominantly

in complex, multi-morbid older individuals with several prior fractures. Despite that, teriparatide remains effective for a wide range of individuals including those with inflammatory arthritis and/or concurrent steroid use. This is in line with recent meta-analysis of real life teriparatide use in complex osteoporosis with multimorbidity. Though NICE stipulates minimum two fractures requirement (unless T score <-4.0), in practice patients may have had more suggesting that clinicians are reserving it for patient with a high disease burden. Our study should enhance clinicians' confidence in teriparatide prescribing. We believe that since the availability of more cost-effective teriparatide biosimilars, NICE ought to amend its guidance and provide easier access to anabolic therapies for better patient outcomes.

Disclosure

R. Vinda: None. **M. Afzal:** None. **M.K. Nisar:** None.